

ASTRAQ

Autonomous Spatial Tracking & Alignment for Optical Links



User Manual

A plain-language guide to every feature of the prototype · Final version

SMART INDIA HACKATHON 2026 · SIH26169

AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile FSOC Terminals
ISRO / Department of Space · Software · Smart Automation, Space Technology

Contents

What is ASTRAQ?	3
Opening ASTRAQ	4
A tour of the screen	5
The tracking journey: what the states mean	7
The five views	8
The sensor panel: what the camera sees	11
The alignment chain and bottom panels	14
The settings drawers	15
Demo mode: the 90-second guided tour	18
Analysis: live graphs and the pass/fail card	19
Experiments: record, export, replay and batch	21
Ready-made scenarios (presets)	22
The video test (camera bypass)	23
Space, themes, zoom and the desktop app	25
Advanced: the Python engine	27
Keyboard shortcuts	28
Troubleshooting and FAQ	29
Glossary	31

Who this manual is for

This manual is written for anyone: a judge, a teacher, a student or a curious visitor. You do not need to know anything about lasers, satellites or control systems. Words that may be new are explained the first time they appear, and there is a glossary at the end (Chapter 18).

CHAPTER 1

What is ASTRAQ?

ASTRAQ is a computer simulation of a truck-mounted laser communication station finding and following a satellite, completely on its own.

Laser links can send huge amounts of data, but the laser beam is extremely thin. For two stations to talk, they must point at each other almost perfectly, even when one of them is on a moving, shaking truck and the other is a satellite racing across the sky at 7.5 km every second.

Before the laser can be switched on, a camera has to **find** a small guide light (the **beacon**) on the satellite and a motorised mount has to turn the camera until the light sits exactly in the middle of the picture. This first step is called **coarse alignment**, and it is what ASTRAQ does.

How it works, in one sentence per step:

Step	What happens	Everyday comparison
1. Look	The camera takes 30 pictures every second.	Looking through binoculars.
2. Find	The software searches each picture for the beacon's bright dot.	Spotting a torch in the dark.
3. Predict	It works out where the dot is moving next.	Guessing where a thrown ball will land.
4. Turn	It turns the mount so the dot moves to the centre.	Following a bird with your binoculars.
5. Hold	Once the dot stays near the centre, the link is LOCKED .	Keeping the bird steady in view.

Everything you see happens for real inside the simulation: the camera picture is actually generated, the dot is actually searched for, and the mount actually turns with realistic speed limits. Nothing is pre-recorded or faked.

Honest scope

ASTRAQ is a **simulation**. It does not control real hardware. Its AI part is a small trained neural network, the **beacon verifier** (Chapter 6), which was trained on simulated pictures only. The "YOLO" detector shown in the app is marked *planned*: it is a place prepared for a future model, not a working feature. The link-budget and acquisition-probability panels are simplified estimates and are labelled that way.

CHAPTER 2

Opening ASTRAQ

There are three ways to open ASTRAQ. The first two need nothing installed.

Way 1: open the web link (easiest)

- Open the ASTRAQ link your team shared (for example `https://astraq...vercel.app`) in **Chrome**, **Edge**, Firefox or Safari on a laptop or desktop.
- Wait about 5 seconds while the Earth is prepared.
- The simulation starts by itself. Within about 1–2 seconds the badge at the top turns green and says **LOCKED**. That is it.

Tip

For the smoothest picture, use Chrome or Edge with "hardware acceleration" switched on (Chrome ► Settings ► System). On an older laptop, choose the **View** drawer ► Render quality ► **Low**.

Way 2: the Windows desktop app

- Unzip ASTRAQ-Windows-x64.zip anywhere (for example on the Desktop).
- Open the folder and double-click **ASTRAQ.exe**. No internet, Node.js or Python is needed.
- If Windows shows "*Windows protected your PC*", click **More info** ► **Run anyway**. This appears because the app is not code-signed.

Way 3: run it from the source code

You need **Node.js version 20 or newer** (free from nodejs.org). Unzip ASTRAQ, open a terminal in the ASTRAQ folder and type:

```
cd web
npm install # only the first time, takes 1–2 minutes
npm run dev # then open http://localhost:5173
```

On Windows you can also double-click `start-web.bat`; on Mac or Linux, run `./start-web.sh`. To stop, press **Ctrl + C** in the terminal.

Optional: the Python engine

ASTRAQ also includes a Python version of the same engine, used for the video test and the programming interface. It is **not needed** for the normal demo. Chapter 13 explains how to start and connect it.

CHAPTER 3

A tour of the screen

The 3D view fills the whole window. Everything else floats on top of it and can be opened or closed.



The main screen. Each number is explained in the table below.

#	Part	What it shows or does
1	Scenario chip	The name of the current scenario (for example "Open Sky").
2	Mission clock	Time since the run started, written T+ minutes:seconds.
3	Engine chip	Which engine is running the simulation ("Local engine" in your browser, or "FastAPI engine" on a server), with its speed in frames per second.
4	State badge	The tracker's current step: SEARCHING, DETECTED, ACQUIRING, TRACKING, LOCKED , LOST or REACQUIRING. The number shows how long it has been in that state.
5	Main buttons	Pause/Run, Reset (restart with a new random start), Record , Run demo , and ? for help.
6	View switcher	Five camera angles: Overview, Terminal, Spacecraft, Sensor POV, Orbit (Chapter 5), and the – / + zoom buttons.
7	Tool rail	Nine icons: the settings drawers, the video benchmark and the measuring tool (Chapter 8).
8	3D scene	Earth, the truck, the satellite, the camera's viewing cone and the red laser link . Drag to look around.
9	Sensor panel	The real picture the camera sees, with the detection marks, or the 2000 × 2000 screen view (Chapter 6).

#	Part	What it shows or does
10	Mission timeline	When each stage (Search → Detect → Acquire → Track → Lock) was first reached, and why the last change happened.
11	Alignment chain	The whole pointing process as a live chain of numbers (Chapter 7).
12	Target panel	Where the satellite's beacon is: direction, distance and speed.
13	Camera panel	Where the mount is pointing, how fast it is turning, and the camera's field of view.
14	Optical link panel	A simplified estimate of laser signal strength, plus the Alignment Quality Score (0-100).
15	Analysis / Experiment	Open the live graphs (Chapter 10) or the Experiment drawer (Chapter 11).
16	Theme	Change the colours of the whole app (Chapter 14).

Moving around the 3D scene

Mouse action	Result
Left-drag	Rotate the view around
Scroll wheel	Zoom in and out, towards the point under the mouse
Right-drag	Slide the view sideways
+ / – keys or buttons	Zoom in and out in steps

CHAPTER 4

The tracking journey: what the states mean

The badge at the top tells you exactly what the tracker is doing. Here is the journey from "where is it?" to "got it".

State	Plain meaning	What you will see
SEARCHING	"I can't see the beacon yet, so I'm scanning the sky."	The mount sweeps in a square spiral. The camera may widen its view to 12° to cover more sky.
DETECTED	"I think I saw it. Let me check it wasn't noise."	It must see the dot in at least 2 of 3 pictures, in the same place, before it believes it.
ACQUIRING	"Confirmed. I'm turning towards it."	The mount swings to bring the dot to the centre. The camera zooms back to its normal 4° view.
TRACKING	"I'm following it."	The dot is being followed but is not yet steady enough.
LOCKED	"It's centred and steady."	The dot has stayed within 10 pixels of the centre for 6 pictures in a row. The badge turns green and the laser link line lights up.
LOST	"I've lost it."	The dot was missing for more than 5 pictures (for example in thick fog).
REACQUIRING	"It should be about here, so I'll look there."	For 1 second the mount keeps following where the dot is predicted to be (the beacon is often only briefly hidden, for example by a cloud). Then it searches in a small spiral around that point. If this fails for 3 seconds, it goes back to SEARCHING.

Tip

Every change of state comes with a reason, for example "**LOCKED**: error < 10 px for 6 frames" or "**LOST**: 6 consecutive misses (coast limit 5)". You can read the latest one under the mission timeline, and the full history in **Experiment ► State transition log**.

While a few pictures are missing, the tracker does not panic. It **coasts**: it keeps turning the mount the way it predicts the satellite is moving. Only after more than 5 missing pictures does it say LOST.

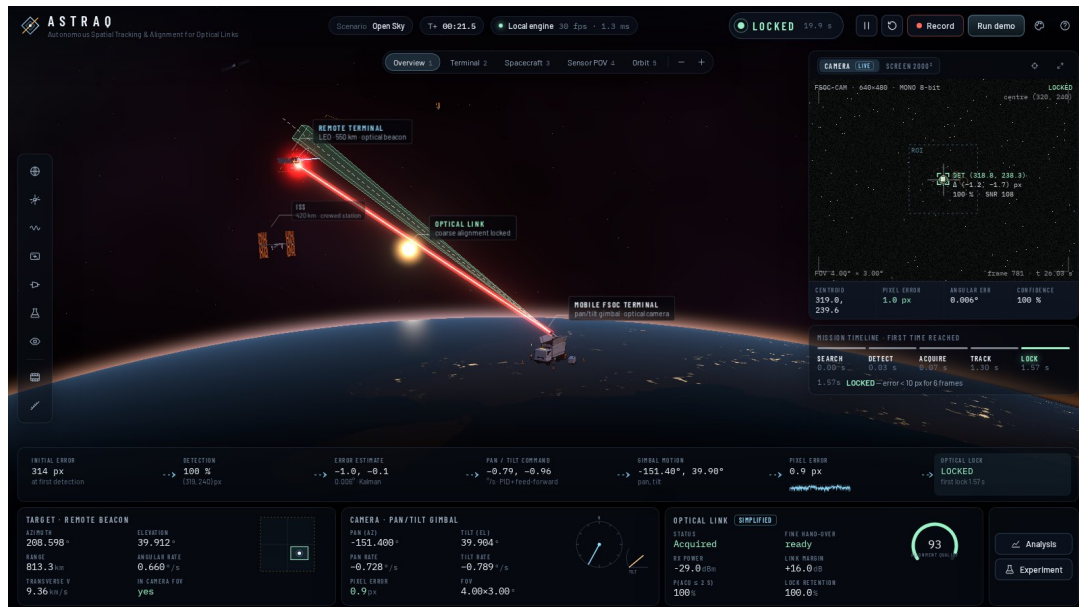
CHAPTER 5

The five views

Switch views with the buttons at the top centre, or press the keys 1 to 5.

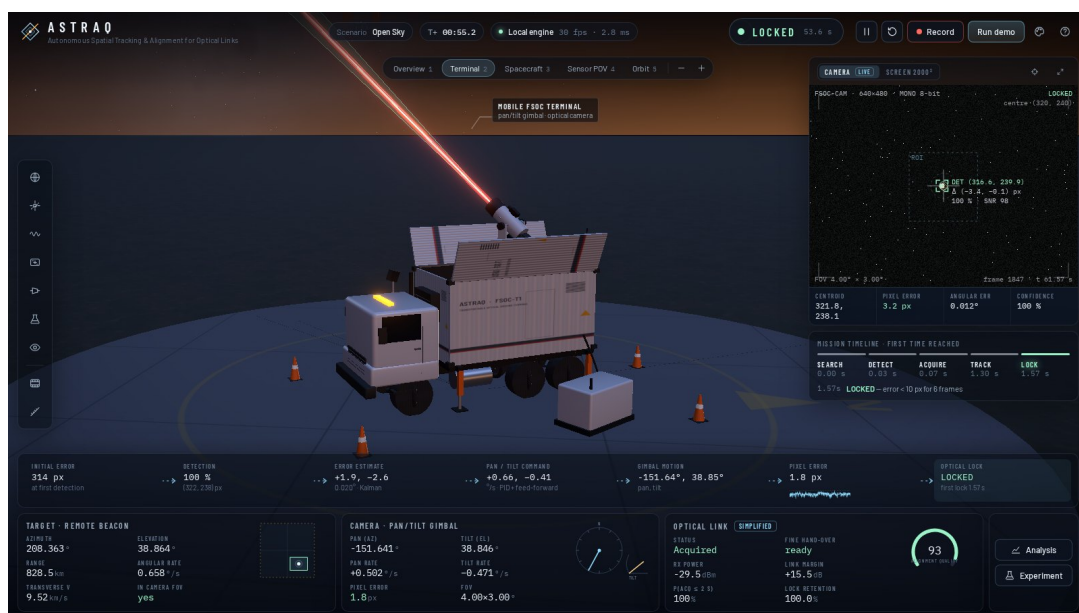
1 • Overview

The whole story at once: the truck on the Earth, the satellite, and the laser link between them.



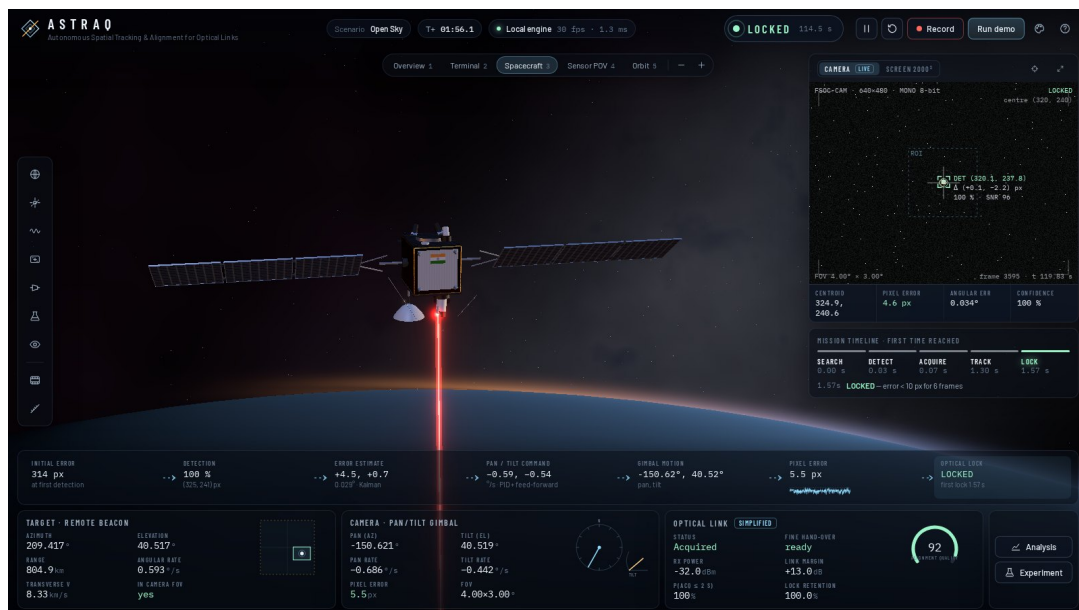
2 • Terminal

A close-up of the **transportable optical ground station**: a six-wheel truck carrying an equipment shelter whose roof panels open. Inside stands a pier with the pan/tilt mount (it turns sideways and up/down), the telescope and the small acquisition camera. Real stations of this kind exist (for example Cailabs TILBA-TOGS and DLR's TOGS). You can see the mount move as it tracks.



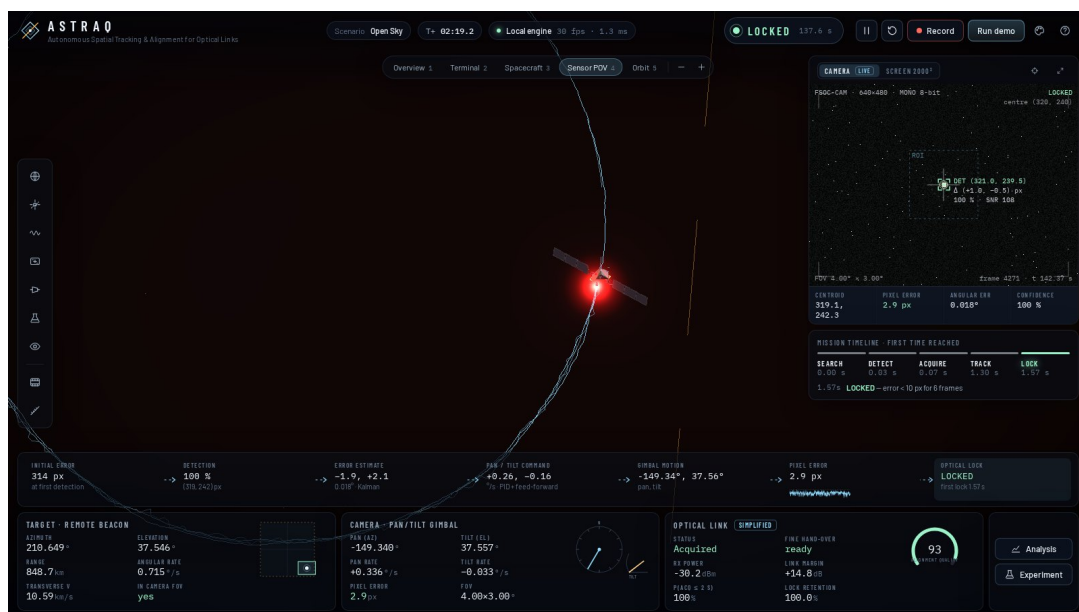
3 • Spacecraft

A close-up of the satellite: a gold-foil body, radiators, two solar wings that turn to face the Sun, antennas, sensors, a flag, and the optical head with its **red beacon** light. **Drag** to look around it and **scroll** to zoom in; the camera travels with the satellite.



4 • Sensor POV

"Point of view": the 3D world as seen through the terminal's telescope, with the camera's real narrow field of view.



5 · Orbit

The big picture from space: the satellite's path over the Earth. Coastlines are real map data.



Zoomed in on the terminal: the opened shelter roof, the pier and the telescope.

Good to know

Directions and distances are true. To make things visible, the truck and satellite are drawn bigger than real when you are far away, the Moon is drawn 3× larger, and the beacon's light cone is widened. The app labels these exaggerations.

CHAPTER 6

The sensor panel: what the camera sees

This is the most important panel. It shows the **actual picture** the detector analyses, 640 × 480 pixels in black and white, 30 times a second.



The sensor panel with its markings.

Mark	What it is
a	Two tabs: CAMERA (with the LIVE tag: the real simulated image) and SCREEN 2000² (see below). Press G to switch.
b	Two buttons: the crosshair toggles a calibration grid (true degree lines), and the arrows enlarge the panel (or press S).
c	ROI ("region of interest"): while tracking, the detector only searches inside this dashed box around the predicted position. This is faster and ignores stars and fake lights elsewhere.
d	The detection box around the beacon, the cross marking its exact centre (the "centroid"), and a small diamond showing where the Kalman filter predicted the beacon would be. A dashed line from the image centre shows the remaining error. A thin ellipse shows how uncertain the prediction is.
e	Readout next to the detection: its pixel position, the error Δ from the centre, the confidence (%) and the signal-to-noise ratio (SNR, how much brighter the dot is than the noise).
f	The current field of view, e.g. 4.00° × 3.00°. During a wide search it shows about 12°.
g	Centroid : the exact pixel position of the dot's centre, down to fractions of a pixel.

Mark	What it is
h	Pixel error: how far the dot is from the image centre (the target for LOCKED is under 10 pixels).
i	Angular error: the same error in degrees. One pixel is 0.00625°.
j	Confidence: how sure the detector is that the dot is really the beacon.

The small white specks are **real stars** from the same star catalogue used for the 3D sky, and the grainy texture is simulated camera noise. In fog you will see the picture become grey and the dot fade; in rain you will see streaks.

How the dot is found (simple version)

- Remove isolated pure-white or pure-black pixels ("salt & pepper" noise).
- Smooth the picture slightly and measure how bright the background normally is.
- Mark everything clearly brighter than the background, then group touching pixels into blobs.
- For every blob, measure 11 simple properties (size, brightness, shape, sharpness, how it stands out from its surroundings and more).
- The **AI beacon verifier**, a small trained neural network, turns those properties into a probability that the blob is the real beacon. Blobs below 70% are rejected, and so are blobs far from where the beacon should be.
- Take the brightness-weighted centre of the best blob. This is accurate to about 1/20 of a pixel.

The AI beacon verifier

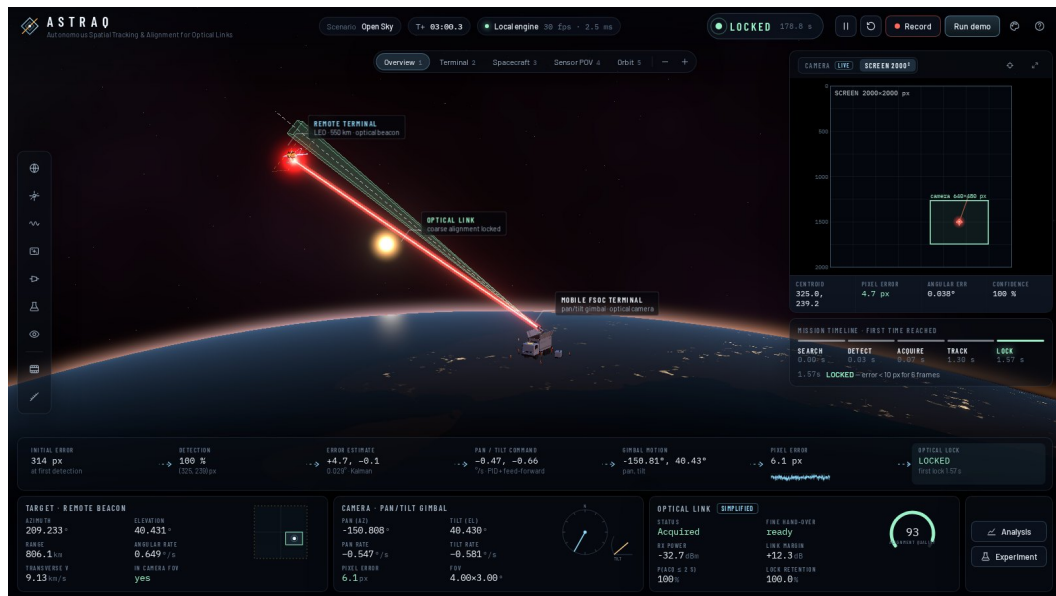
Stars, noise speckles and fake glints can look like the beacon. The verifier learned the difference from thousands of labelled simulated pictures in clear sky, haze, fog, rain and low light. On 3,879 test pictures it had never seen, it found the beacon in **98.7%** of them (the older hand-tuned rule: 89.9%) and accepted a wrong spot in only **0.6%** (hand-tuned: 8.4%). In the Weak Beacon scenario the time to lock fell from about 4.6 s to 1.4 s.

Tip

Compare it yourself: Tracking drawer ► Detection provider ► switch **Learned beacon verifier (AI)** off and on, with the Weak Beacon preset.

The screen view (2000 × 2000)

Click **SCREEN 2000²** at the top of the sensor panel (or press **G**). It shows where the beacon is on a 2000 × 2000 display, like the full-screen video used in the problem statement's second test.

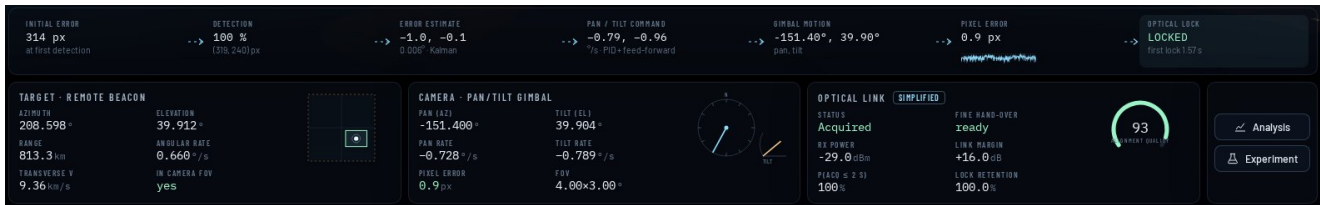


The screen view.

CHAPTER 7

The alignment chain and bottom panels

The strip at the bottom shows the whole pointing process as a chain, from left to right, with live numbers.



The alignment chain (top row) and the Target, Camera and Optical link panels.

Box in the chain	Meaning
Initial error	How far off the beacon was when first detected (in pixels).
Detection	Detector confidence and the pixel where the dot was found.
Error estimate	The filtered error after the Kalman filter has smoothed out noise.
Pan / tilt command	How fast the controller is telling the mount to turn (degrees per second), sideways (pan) and up/down (tilt).
Gimbal motion	Where the mount is actually pointing now (pan and tilt angles).
Pixel error	The resulting error in the picture, with a tiny live graph.
Optical lock	LOCKED or not, and when the first lock happened.

The three panels underneath

- **Target · Remote beacon:** the beacon's direction (azimuth = compass direction, elevation = height above the horizon), distance in km, how fast it moves across the sky, and whether it is inside the camera's view. A mini-map shows the beacon inside the search field.
- **Camera · Pan/tilt gimbal:** the mount's angles and turning speeds, the pixel error, the field of view, and two small dials showing pan and tilt.
- **Optical link** (marked *simplified*): an estimate of the received laser power and the safety margin, whether the error is small enough to hand over to a "fine" pointing system, the chance of acquiring within 2 s, lock retention, and the **Alignment Quality Score** gauge.

What is the score?

Alignment Quality Score (AQS) is a simple 0–100 health score over the last 10 seconds: 45% from how small the error is, 25% from detection confidence and 30% from the time spent locked. It is a plain formula, not AI.

CHAPTER 8

The settings drawers

The icons on the left open drawers. Every change applies immediately, while the simulation keeps running.



The tool rail, from top to bottom.

Tip

If you change too much and the tracker struggles, open **Scenario** and click the **Open Sky** preset to go back to the defaults.

Scenario

Section	What you can do
Scenario presets	One click loads a complete, ready-made test situation (see Chapter 12).
Run	The random "seed" (the same seed always gives the same run, which is useful for fair comparisons) and the time speed.
Geometry	Who the partner is (LEO satellite, HAPS high-altitude platform, or UAV drone), satellite altitude, the highest point of its pass across the sky, the ephemeris error (how wrong the predicted position is, in seconds), the predicted direction, and the Sun's position for lighting.

Target

Section	What you can do
Trajectory	How the beacon moves: stationary, linear, circular, sinusoidal (wavy), figure-8, spiral, random, custom (click points to draw your own path) or LEO pass (a real satellite orbit). Set speed, size (amplitude), period (time for one loop) and direction.
Start position	Random (default) or a fixed start. Applies when you press Reset.
Beacon spot	Shape (square, disk or soft Gaussian), size 5–20 pixels, and brightness.

Disturbances

This is where you make life hard for the tracker, like real weather and a bumpy road.

Section	Controls and meaning
Atmosphere	Clear, haze, fog, rain, low light , plus a strength slider (how much contrast and brightness is lost) and turbulence (twinkling and wandering of the beacon, like heat shimmer).
Sensor noise	Gaussian (fine grain, up to $\sigma = 20$), salt & pepper (random white and black pixels, up to 10%) and Poisson (the natural graininess of light).
Platform & gimbal	Vibration (engine shake: size and frequency), camera jitter (random shake up to ± 20 pixels per picture), platform motion (the truck swaying: linear, circular or random) and wind torque (gusts pushing the mount).
Target & detection	Target motion noise (the beacon wobbles), beacon dropouts (the dot disappears in some pictures), cloud outage (the beacon is hidden for a set time, every so many seconds) and a decoy glint (a second fake bright light to try to fool the tracker).

Tracking

Section	Controls and meaning
Pointing mode	Automatic tracking or Manual pan/tilt (you steer with the arrow keys). Force reacquisition makes the tracker search again.
Detection provider	Image centroid + AI verifier : the real image-processing detector (default), with the Learned beacon verifier (AI) switch. Synthetic : a labelled test model that uses the true position plus added noise, for studying the controller on its own. YOLO : shown as <i>planned</i> , not available. Sliders set the brightness threshold, minimum confidence, how far from the prediction a detection may be (association gate), and extra delay or miss chance.
Kalman filter	Switch the prediction filter on or off, and tune how jumpy the target is expected to be (process noise q) and how noisy measurements are (measurement noise r). Try switching it off to see tracking get worse.
PID controller	The steering "gains" K_p , K_i and K_d , the integral limit, the control rate, and rate feed-forward (use the predicted speed to turn in advance).
Gimbal limits	Maximum turning speed (the problem statement allows 5–10°/s), maximum acceleration, and the tilt range.
State machine thresholds	How close counts as locked (default 10 px), how many pictures in a row are needed, how many missed pictures to coast through, and how long to search locally before starting over.

Optics & link

Section	Controls and meaning
Camera calibration	A table and diagram of the camera maths: focal length, pixel size, and degrees per pixel (0.00625°).
Camera	Resolution (640 × 480), field of view (default 4° wide), frame rate (30 per second), and wide-field acquisition : search with a wide 12° view, then zoom in to 4° once found. This makes finding the beacon about 6 times faster.

Section	Controls and meaning
Acquisition probability	An <i>estimate</i> of the chance of finding the beacon within 2 seconds, with its formula shown.
Link budget	A <i>simplified</i> laser power calculation: transmit power, beam spread, receiver size and sensitivity, and the capture range of the fine pointing stage.

Experiment

Recording, replay, batch tests, performance reports and the engine choice. Chapter 11 covers this drawer in detail.

View

Section	Controls and meaning
Theme	The colour theme (Chapter 14).
Render quality	Low (fastest, for older laptops), Medium or High (clouds, glow effects, shadows, sharper edges). If your graphics card cannot draw High, ASTRAQ switches down by itself and tells you.
Space environment	Show or hide the other satellites, the ISS, meteors, aurora and galaxies, and Fly to each spacecraft.
3D overlays	Show or hide the camera's viewing cone and optical axis, the paths and search area, labels, and the latitude/longitude grid.
Sensor overlays	Show or hide the ROI box, the calibration grid, and a ground-truth marker (where the beacon really is; simulation only).
Scale	How much the truck and satellite are enlarged for visibility.

Video benchmark (film icon)

Opens the video test panel (Chapter 13).

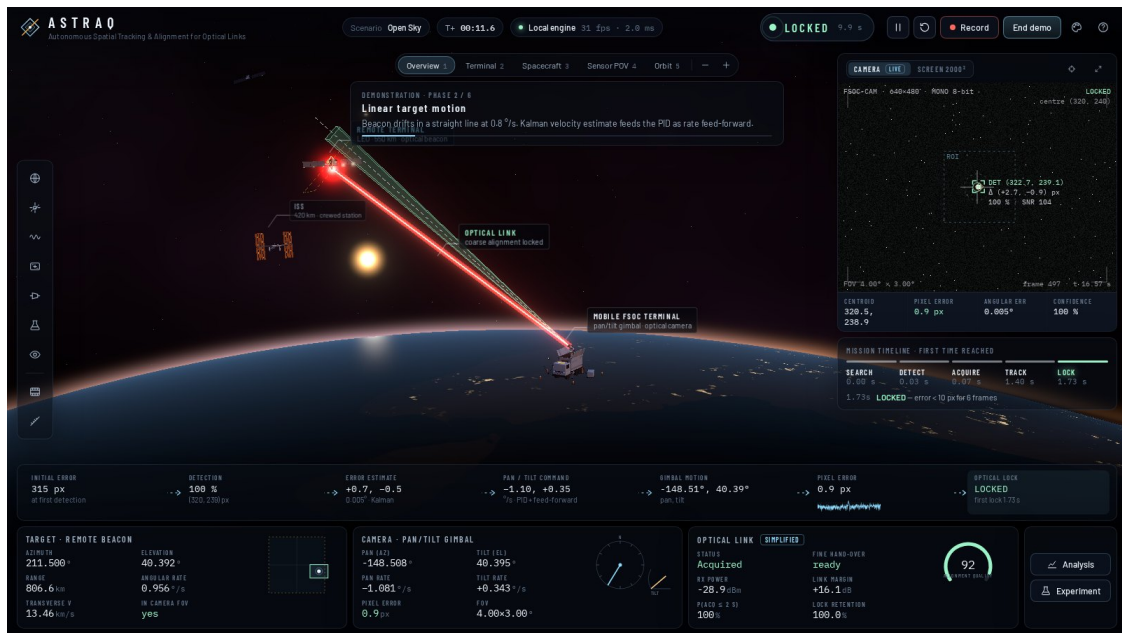
Measure (ruler icon)

Click the ruler, then click the truck, the satellite, the Moon or any point on the Earth. ASTRAQ shows the true distance, the direction (azimuth and elevation) and the angle away from where the camera is pointing. Click a second object to get the distance and angle between the two. Click the ruler again to stop measuring.

CHAPTER 9

Demo mode: the 90-second guided tour

Click **Run demo** (top right) or press **D**. A caption at the top explains each part.



Demo mode with its caption.

Time	Phase	What to watch
0-10 s	Finding a still beacon	The camera widens, spirals, spots the dot, zooms in and locks.
10-25 s	Straight-line motion	The mount glides smoothly, using the predicted speed.
25-45 s	Circular motion	The dot stays inside the 10-pixel lock circle.
45-60 s	Wavy motion	A tougher, faster side-to-side movement.
60-70 s	Bad conditions	Fog, heavy noise and 60% of pictures missing the dot. Watch it coast, go LOST and search nearby.
70-90 s	Recovery	Skies clear, the dot is found again where it was predicted, and the link returns to LOCKED on a figure-8 path.

The demo only changes the **situation** (motion and weather). How the tracker reacts is the real, unscripted response. You can switch views while it runs. Click **End demo** or press **D** again to stop.

CHAPTER 10

Analysis: live graphs and the pass/fail card

Click **Analysis** (bottom right) or press **A**.



The Analysis panel (centre) with the Tracking drawer open on the left.

Six graph tabs

Tab	Shows
Pixel error	The true pointing error and the tracker's own estimate over the last 60 s, with the 10 px lock line.
Angular error	The same error in degrees.
Pan / tilt response	The commanded versus the actual mount movement.
Target rate	How fast the beacon moves across the sky.
Confidence	Detector confidence over time.
Quality score	The Alignment Quality Score over time.

A coloured band under each graph shows which state the tracker was in at each moment.

The PS169 acceptance card

This card checks the current run against the official targets from the problem statement, with a tick when a target is met:

Requirement	Target	Typical ASTRAQ result
Acquisition time	≤ 2 s	about 1.1–1.6 s
Tracking error (RMS)	≤ 10 px	about 3 px
Target loss	$< 5\%$	0%

Requirement	Target	Typical ASTRAQ result
Re-acquisition	≤ 1 s	about 0.03 s after a short outage
Lock retention	(reported)	share of visible time spent locked
Processing speed	≥ 20 FPS	30 or more

Below the card are details such as mean, 95th-percentile and maximum error, and how accurate the centroid is compared with the true dot. Time when the beacon is hidden (for example by a cloud) is not counted as a tracking loss, and re-acquisition time is measured from the moment the beacon is visible again.

Tip

The **Report** button in this panel (or the **P** key) downloads a performance report of the current run (Chapter 11).

CHAPTER 11

Experiments: record, export, replay and batch

Open the **Experiment** drawer (flask icon, or the Experiment button bottom right).

Record and export

- Click **Record** (top bar or drawer). ASTRAQ stores every frame's data: 30 rows per second.
- Click **Stop**.
- **CSV** downloads a spreadsheet (opens in Excel) with one row per frame: time, state, beacon and mount angles, detected and true pixel positions, errors, Kalman estimates, processing time and more.
- **JSON** downloads the complete run, including settings and all events, and can be replayed later.

Replay

Click **Replay** to play your recording back in the same screen, with play, pause and a seek bar. Use **Load JSON** to replay a file saved earlier. The top chip shows "Replay" while this is active.

Batch (many runs at once)

Choose a number of runs and a duration, then start. ASTRAQ runs that many simulations in the background, each with a different random start, and reports the average and worst acquisition time, the error, the loss and how many runs passed. This is how the results in this manual were measured.

Performance report

The problem statement asks for an automatically generated performance report. Click **Report (live run)** or **Report (recording)** in the Experiment drawer, the **Report** button in the Analysis panel or after a batch, or press **P**. The report lists every required number (acquisition time, pointing error, loss, re-acquisition, lock retention, speed, centroid accuracy) with its limit and a pass or fail mark, plus the settings used and the event log. It opens in any browser (HTML) and can also be saved as Markdown or JSON. Switch on **Save a report automatically when a recording stops** to get one after every recording.

State transition log

A list of every state change in the run, with the time and the reason.

Engine

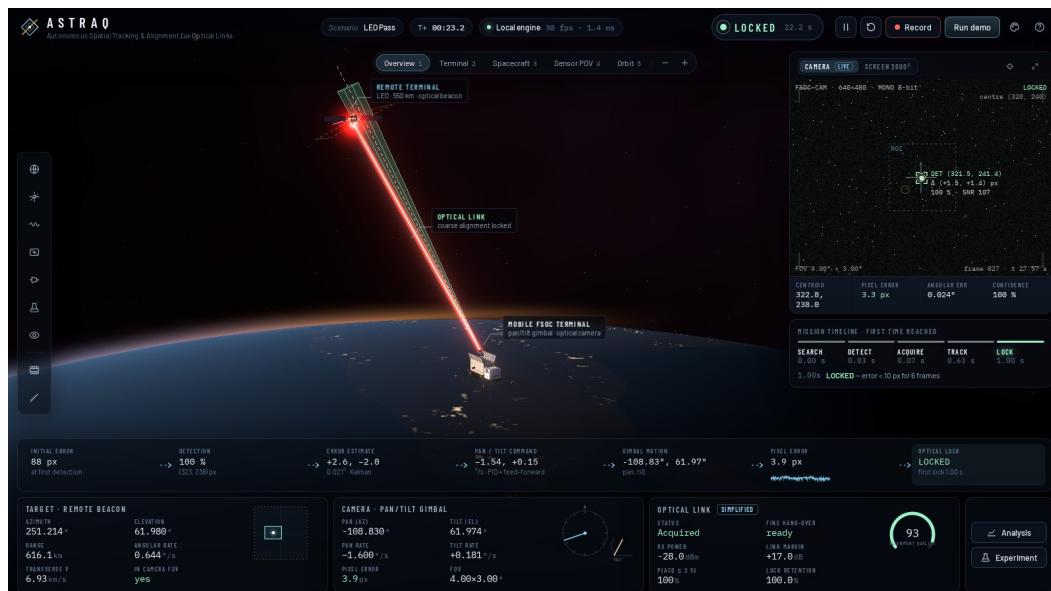
Choose **Local (browser)** (the default) or **FastAPI server** (see Chapter 13).

CHAPTER 12

Ready-made scenarios (presets)

Scenario drawer ► Scenario presets. One click sets up a whole test.

Preset	What it tests	Result over 20 random runs
Open Sky	Clear sky, normal settings (the default)	All locked; about 1.1 s to lock; about 3 px error
PS169 Baseline	Only the official 4° camera, no wide search	All locked, but slower to find (average 6.7 s), because the narrow camera has to scan more
Moving Platform	The truck sways	All locked; about 7 px error
High Jitter	Maximum camera shake (± 20 px per picture)	All locked; about 17 px error. Shake this strong cannot be removed by any mount, so this is expected.
Weak Beacon	Haze, a dim, small dot, heavy noise	All locked in about 1.3 s (about 4 s without the AI verifier)
Fast Target	A quickly moving beacon and a faster mount	All locked; about 10 px error
Acquisition Challenge	Starts in a corner, with rain, 30% missing pictures and a fake light	All locked; about 3.9 s
Occlusion & Reacquisition	The beacon is hidden for 1 s every 6 s, like passing clouds	All locked; every outage recovered 0.03 s after the beacon reappears
LEO Pass	A real satellite orbit with a timing error	All locked in about 1 s; about 3 px error

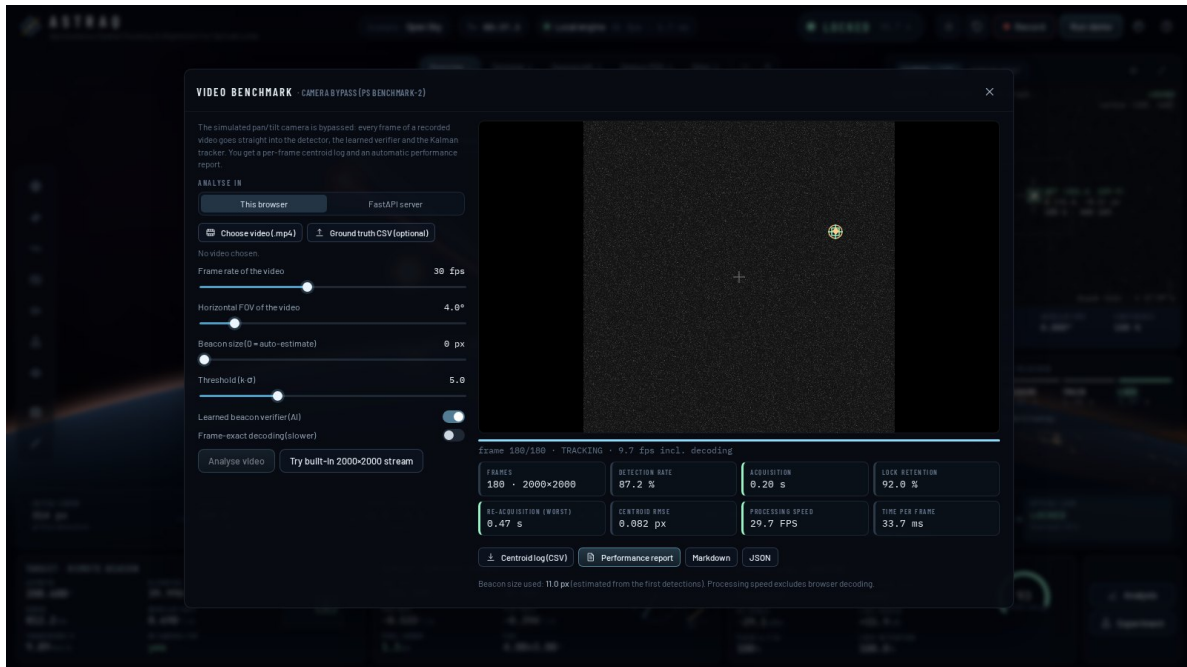


The LEO Pass preset.

CHAPTER 13

The video test (camera bypass)

The problem statement's second test gives a video file instead of the simulated camera. ASTRAQ can analyse it right in the app.



The video benchmark panel after analysing the built-in 2000 × 2000 test video.

- Click the **film icon** on the tool rail, or press **V**.
- Choose **Analyse in: This browser** (no server needed) or **FastAPI server** (Chapter 15).
- Click **Choose video (.mp4)** and pick an MP4 or WebM file. Optionally add a **truth CSV** with the real beacon position in each frame (columns frame, x, y); then ASTRAQ can calculate the error.
- No video at hand? Click **Try built-in 2000×2000 stream**: a generated full-screen video with heavy speckle noise, a moving beacon and a short outage, with the exact truth.
- Set the camera's field of view if you know it. Leave **Beacon size** at 0 and ASTRAQ measures it automatically.
- Click **Analyse video**. You see each frame with the detection, then the results.

Result	Meaning
Detection rate	Share of frames where the beacon was found.
Acquisition	Time until the tracker first locked on.
Lock retention	Share of the video spent locked.
Re-acquisition (worst)	Longest time to find the beacon again after it came back.
Centroid RMSE	Average distance between the found centre and the true centre, in pixels (needs a truth CSV).
Processing speed	Frames analysed per second (the target is at least 20).

Download the per-frame **centroid log** (CSV) or the **report**. Test results: on the built-in 2000 × 2000 video the error was **0.08 pixels** at 30 frames per second, and the tracker found the

beacon again 0.47 s after the outage.

Good to know

Some browsers cannot play every MP4 file. Chrome and Edge on Windows or Mac play normal MP4 files. If a file will not load, convert it to WebM or use the FastAPI server option.

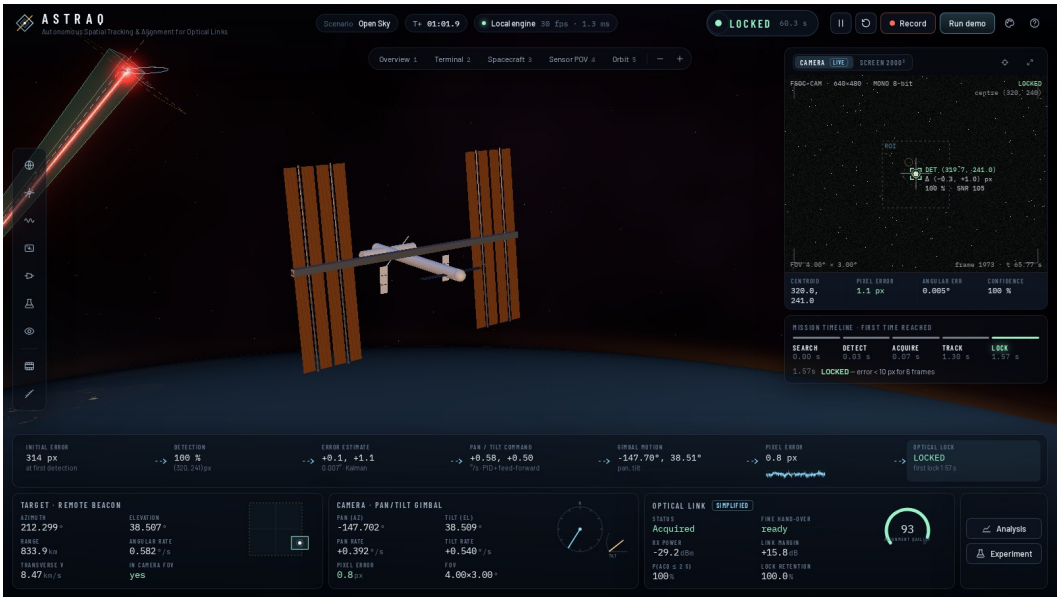
CHAPTER 14

Space, themes, zoom and the desktop app

Explore the space around the link, and make ASTRAQ look right for the room you are presenting in.

The space around the link

Real near-Earth space is busy, so ASTRAQ shows it: two more satellites (**SAT-2**, an Earth-observation satellite at 700 km, and **SAT-3**, a data-relay satellite at 1,100 km), the **International Space Station** at 420 km, **meteors** streaking through the upper atmosphere, the green **aurora** around the poles, and the **Andromeda galaxy** and **Magellanic Clouds** in the sky. The spacecraft move at their real orbital speed.



Fly to ► ISS: the chase camera beside the International Space Station.

- Press **F** (or View drawer ► Space environment ► **Fly to**) to jump beside the ISS. Press **F** again for SAT-2 and SAT-3. Press **1** to go back.
- **Look around any spacecraft:** drag with the mouse to move around it, scroll to zoom in or out. The camera travels with the spacecraft, so it never flies away. This also works for the main satellite in the **Spacecraft** view (key 3).
- Turn all of it on or off in View drawer ► **Space environment**.

Good to know

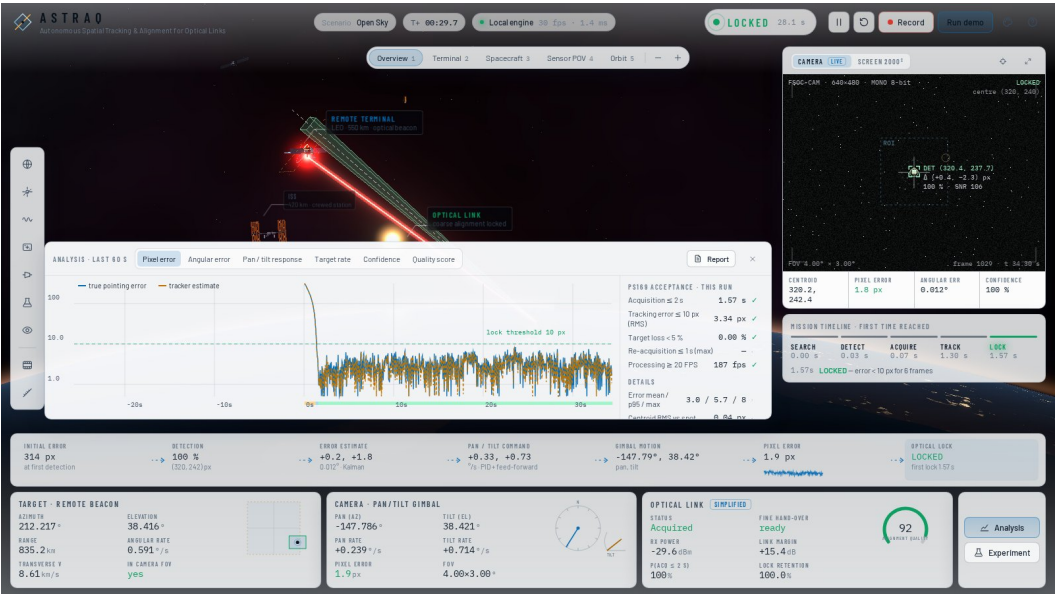
These objects are only for the picture. They are not in the camera image the tracker analyses, so they cannot change tracking or any result. The galaxies are at approximate positions.

Colour themes

Click the **palette** button in the top bar (or press **T**, or View drawer ► Theme). Five themes are available:

Theme	Best for
Deep Space (default)	Dark rooms and screens; navy with ice-blue highlights.
Laser Red	Matches the red laser beam; charcoal with crimson.

Theme	Best for
Aurora	Dark teal with mint.
Solar Gold	Graphite with amber.
Daylight	Bright rooms and projectors : light panels with dark text.



The Daylight theme.

ASTRAQ remembers your theme the next time you open it.

Zooming in closely

The scroll wheel zooms towards whatever is under the mouse, so point at the truck or the satellite and scroll to fly in. The + and – keys, or the small – / + buttons next to the view switcher, zoom in steps.

The desktop app

ASTRAQ.exe is the same app packaged for Windows. It runs without internet, which is useful when the venue network is unreliable. Chapter 2 explains how to start it.

CHAPTER 15

Advanced: the Python engine

For technical users. The normal demo never needs this.

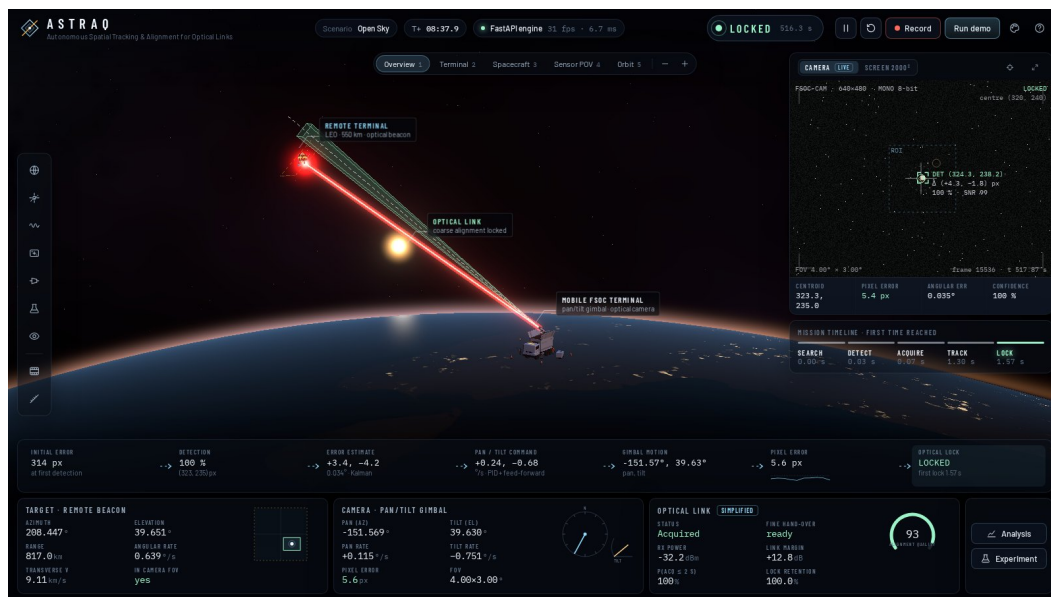
Start the server

```
cd server
python -m venv .venv
source .venv/bin/activate (Windows: .venv\Scripts\activate)
pip install -r requirements.txt (first time only)
python -m uvicorn app.main:app --port 8000
```

Open <http://localhost:8000/docs> to see and try every command of the programming interface (API) in your browser.

Connect ASTRAQ to it

Experiment drawer ► Engine ► choose **FastAPI server** ► check the address ► **Connect**. The chip at the top changes to **FastAPI engine**, and the picture and numbers now come from the Python server. If the server cannot be reached, ASTRAQ tells you and keeps using the browser engine.



ASTRAQ running on the Python FastAPI engine.

The video test (camera bypass)

Besides the video panel (Chapter 13), you can send an MP4 file to `/api/video/analyze` (easiest from the `/docs` page, using "Try it out"). Test results: 0.06 pixels average error at 280 frames per second on 640 × 480 video, and 0.12 pixels at 32 frames per second on 2000 × 2000 video with speckle noise. Reports are available at `/api/report`.

```
python scripts/make_test_video.py test.mp4 8 # makes a test video + test_truth.csv
python scripts/make_test_video.py big.mp4 6 --size 2000 --sp 0.01 --outage 3:3.6
curl -F file=@test.mp4 -F truth=@test_truth.csv http://localhost:8000/api/video/analyze
```

CHAPTER 16

Keyboard shortcuts

Press **?** in the app at any time to see these.

Key	Action
Space	Pause / run
R	Reset: new random start
D	Start / stop the 90 s demo
1 2 3 4 5	Views: Overview · Terminal · Spacecraft · Sensor POV · Orbit
S	Enlarge / shrink the sensor picture
A	Open / close the Analysis graphs
G	Sensor panel: camera picture / 2000 × 2000 screen view
V	Video benchmark
P	Download the performance report
T	Colour themes
+ / −	Zoom in / out
F	Fly to the ISS, SAT-2 or SAT-3 (press again for the next); drag to look around, scroll to zoom
M	Manual steering mode on/off
← ↑ → ↓	In manual mode: nudge the mount 0.1° (hold Shift for 1°)
?	Help overlay
Esc	Close the open drawer or panel

Try it: steer it yourself

Press **M**, then use the arrow keys to move the mount. Watch the dot move in the sensor picture. Press **M** again and the automatic tracker takes back control and locks on again.

CHAPTER 17

Troubleshooting and FAQ

Problem	What to do
The 3D area is black or empty	Wait 5 seconds. If it happens at High quality, ASTRAQ now notices and steps down to Medium by itself, with a message. Otherwise try View ► Render quality ► Medium or Low, and turn on hardware acceleration in the browser settings. Remote-desktop sessions often cannot show 3D.
"WebGL is not available"	The browser or graphics card has 3D switched off. Use an up-to-date Chrome or Edge with hardware acceleration on.
It feels slow or jerky	Choose Low quality, close other heavy tabs, or make the window smaller. The simulation itself still runs at full speed.
It never locks	The conditions may be too extreme (for example very thick fog plus maximum noise). Click the Open Sky preset to reset everything.
"npm is not recognized"	Install Node.js 20 or newer from nodejs.org, then close and reopen the terminal.
Port already in use	ASTRAQ is already running in another window. Close it, or run <code>npm run dev -- --port 5174</code> .
A video will not load	The browser cannot play that file type. Use Chrome or Edge, convert the video to WebM, or choose FastAPI server in the video panel.
Windows blocks ASTRAQ.exe	Click More info ► Run anyway. The warning appears because the app is not code-signed.
Cannot connect to the FastAPI server	Start the server first and check that <code>http://localhost:8000/api/health</code> opens. For an online server that was idle, wait about a minute and try again.

Frequently asked questions

Is the camera picture real or drawn?

It is generated by the simulation from the camera's exact position, including stars, weather and noise, and the detector really analyses it.

Where is the "AI"?

A trained neural network, the beacon verifier, decides which bright spot is really the beacon (Chapter 6). It made detection far more reliable and cut false detections by more than ten times. Around it, the system makes its own decisions: it detects, predicts, steers, and decides when it has lost or found the target, with reasons. It was trained on simulated pictures; a YOLO detector is prepared but not included, and the app says so.

Why is the ground station a truck?

Real transportable laser ground stations are built this way: a shelter or container with an opening roof that is driven to where it is needed. ASTRAQ's model follows that design.

Why does the camera zoom out at the start?

A 4° camera would take up to 12 seconds to scan the whole search area. Scanning with a 12° view first and then zooming in finds the beacon in about 1 second. Real systems such as NASA's OPALS used the same idea.

Why is High Jitter above 10 pixels?

Random shaking of ± 20 pixels in every picture is, by itself, about 16 pixels of error on average. A motorised mount cannot cancel shaking that changes direction faster than it can see it. On real terminals a second, faster "fine" stage handles this.

Do I need the Python server?

No. Even the video test runs in the browser. The server is an optional second engine with a programming interface.

Does it control real hardware?

No. It is a simulation. The code has clear places where a real camera and mount could be connected in future.

CHAPTER 18

Glossary

Word	Simple meaning
Acquisition	Finding the beacon and bringing it to the centre for the first time.
Alignment	Pointing two things exactly at each other.
Azimuth	Compass direction (0° = north, 90° = east).
Beacon	A bright guide light on the partner station that the camera looks for.
Centroid	The exact centre of the bright dot, weighted by brightness.
Coarse alignment	The first, rough stage of pointing: camera plus motorised mount.
Confidence	How sure the detector is that it found the real beacon.
Elevation	Angle above the horizon (0° = horizon, 90° = straight up).
Ephemeris	The predicted position of a satellite over time.
Feed-forward	Turning in advance using the predicted speed, so the mount does not lag behind.
Field of view (FOV)	How wide an area the camera sees, in degrees.
FPS	Frames (pictures) per second.
FSOC	Free-Space Optical Communication: sending data with light through air or space.
Gimbal / pan-tilt mount	A motorised mount that turns sideways (pan) and up/down (tilt).
Jitter	Small, fast, random shaking of the camera.
Kalman filter	A method that combines noisy measurements with a motion guess to estimate where something is and how fast it moves.
LEO	Low Earth Orbit, a few hundred km up (550 km here).
Lock	The beacon has stayed near the centre long enough to be considered aligned.
Noise	Random speckles and grain in a camera picture.
Occlusion / outage	A time when the beacon is hidden, for example by a cloud.
PAT	Pointing, Acquisition and Tracking: the whole process of keeping a laser link aligned.
PID controller	A standard steering method that turns faster when the error is big and gently when it is small.
Pixel	One dot of the camera picture. Here one pixel = 0.00625° .
Re-acquisition	Finding the beacon again after it was lost or hidden.
RMS error	A kind of average error that gives extra weight to big mistakes.
ROI	Region of interest: the part of the picture being searched.
Seed	A number that fixes the random choices, so a run can be repeated exactly.

Word	Simple meaning
SNR	Signal-to-noise ratio: how much the dot stands out from the noise.
Verifier (AI)	A small trained neural network that checks whether a bright spot is really the beacon.